

HOSPITAL EMERGENCY ROOM DASHBOARD

Power BI Analytics Project

Patient Management • Wait Time Analysis • Satisfaction Insights • Referral Tracking

April 2023 – October 2024 | 9,216 Unique Patients



Steps in Project

01

Requirement Gathering

Define business goals and KPIs for the ER dashboard

02

Data Walkthrough

Understand dataset structure, fields, and relationships

03

Data Connection

Connect Power BI to the Excel data source

04

Data Cleaning

Handle missing values, duplicates, and quality checks

05

Data Modeling

Build relationships between Date Table and ER Data

06

DAX Calculations

Create measures: No. of Patients, Avg Wait Time, etc.

07

Dashboard Layout

Design wireframe and visual structure for 4 dashboards

08

Charts & Formatting

Develop all visuals with consistent styling

09

Insights Generation

Write key takeaways and descriptive analysis



Dashboard Overview — 4 Views

Dashboard 1: Monthly View

Monitor KPIs and trends on a month-by-month basis. Filter by Month Name and Year to track patterns and seasonal changes.

Dashboard 2: Consolidated View

Holistic summary of hospital performance across a customizable date range for broader trend analysis.

Dashboard 3: Patient Details

Granular patient-level data grid with ID, name, gender, age, wait time, referral, and admission status.

Dashboard 4: Key Takeaways

Descriptive analysis with actionable insights, patterns, anomalies and recommendations for stakeholders.



KPI Requirements



9,216

No. of Patients

Total unique patients visiting the ER. Displayed with a daily area sparkline to identify peak days and seasonal trends.



35.3 Min

Avg Wait Time

Average time patients wait before being attended to. Sparkline reveals daily fluctuations and operational pressure points.



4.99 / 10

Patient Satisfaction

Average satisfaction score evaluated daily. Dips in score are correlated with peak hours and operational challenges.



3,816

Patients Referred

Count of patients referred to specialist departments. Tracks departments with high referral volumes needing more resources.



Dashboard 1 — Monthly View: Charts to Build

Table Visual

Patient Admission Status

Admitted vs Not Admitted patient count with % triage breakdown.

Column Chart

No. of Patients by Age Group

Patients grouped in 10-year intervals: 0-9, 10-19 ... 70-79.

Bar Chart

Department Referrals

Referral volume per department: General Practice, Ortho, Physio, Cardiology.

Donut Chart

Timeliness — Within 30 Mins

% of patients seen within 30 minutes: Target Met vs Target Missed.

Donut Chart

No. of Patients by Gender

Male, Female, and Not Confirmed patient distribution.

Bar Chart

Patient Race Distribution

Breakdown by White, African American, Asian, Multi-racial, etc.

Heatmap Matrix

No. of Patients by Day & Hour

Hour rows (00-02 to 23-24) vs Day columns (Mon–Sun) showing volume.



Key DAX Measures & Calculated Columns

Measure	No. of Patients	DISTINCTCOUNT('Hospital_ER_Data'[Patient Id])
Measure	Avg Wait Time	AVERAGE('Hospital_ER_Data'[Patient Waittime])
Measure	Patient Satisfaction Score	AVERAGE('Hospital_ER_Data'[Patient Satisfaction Score])
Measure	No. of Patients Referred	CALCULATE([No. of Patients], 'Hospital_ER_Data'[Department Referral] <> "None")
Column	Age Group	SWITCH(TRUE(), [Patient Age] >= 70, "70-79", [Patient Age] >= 60, "60-69", ... "0-9")
Column	Admission Status	IF([Patient Admission Flag] = TRUE, "Admitted", "Not Admitted")
Column	Waittime Interval	SWITCH(TRUE(), [Admission Time] < 2, "00-02", < 4, "03-04", ... "Above 24")
Column	Month & Year	FORMAT('Date_Table'[Date], "MMM YYYY")

* All measures stored in Hospital_ER_Data table - Date Table used for time intelligence



Key Insights & Takeaways

 **9,216 Patients**

19-month period

High patient volume requiring round-the-clock staffing and resource planning.

 **35.3 Min Avg Wait**

Improvement needed

Above optimal threshold. Monday mornings and late nights are peak pressure periods.

 **4.99 / 10 Score**

Moderate satisfaction

Directly correlated with wait times. Reducing wait time will improve scores significantly.

 **50% Admitted**

4,612 admitted

Nearly equal admitted vs treated-and-released split demands constant inpatient bed availability.

 **Monday Busiest**

1,377 patients

Saturdays and Tuesdays also elevated. Staff allocation must reflect weekly demand patterns.

 **Top Referral: GP**

1,840 cases

General Practice leads, followed by Orthopedics (995). These departments need priority resources.



Things to Watch Out for in Power BI

01

Use Old Card Visual

New Card has 'Categories' field — use the classic Card (only 'Value' field) for single metric display like No. of Patients.

03

Formula Bar Visibility

If formula bar is missing when creating measures, go to View tab and enable Formula Bar. Always select the table first before creating a new measure.

05

Visual Layering

If a card appears behind a shape, use View → Selection pane to drag the card layer above the shape layer.

02

Display Units = None

Power BI auto-formats large numbers as 9K. Go to Format → Callout/Value → Display Units → None to show full number.

04

Correct Column Names in DAX

Always verify column names in the Data pane before writing DAX. Typos like 'Admission Hour' vs 'Admission Time' will cause #ERROR.

06

KPI Visual Needs Trend Axis

KPI visual requires a date field (Month Name) in the Trend Axis field — without it, the value won't display.

Thank You

Hospital Emergency Room Dashboard — Power BI Project

Built with: Power BI Desktop • DAX • Data Modeling • Excel Data Source

April 2023 – October 2024 | 9,216 Patients | 4 Dashboard Views

"Data-driven decisions for better emergency care."